

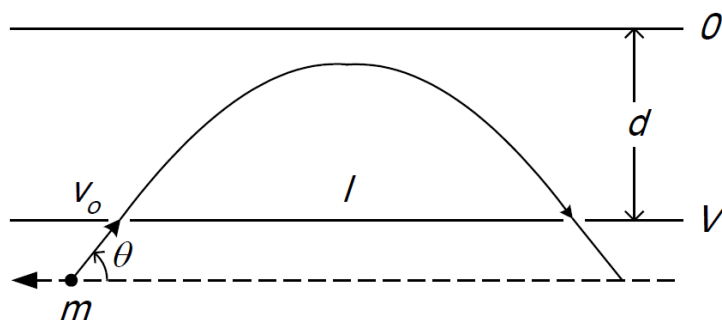
PHYS 5350 Assignment 1

Due date: One week after announcement

Total 100 + 20 points

1. Show that the parallel plate capacitor makes an electron energy analyzer with focusing action when the entrance angle is 45° . The speed of the electron entering the capacitor is v_0 . The separation between the plates is d and the voltage across the plates is V . (40 points)

Hint: Find l in terms of the kinetic energy of the electron at the entrance and show that it does not change for small variations of θ around 45° .



(Optional) Extend the ideas obtained from the parallel plate analyzer to a cylindrical analyzer by showing that the latter also has the capability of analyzing the energy of and focusing the electrons (extra 20 pts).

2. Why is it necessary to mount the source, crystal and the detector on a Rowland circle for an X-ray monochromator? Derive a relationship among the distance between crystal and detector, the radius of the circle, the wavelength of X-ray and the lattice parameter of the crystal (assuming cubic). (40 points)

3. $MgK\alpha$ and $AlK\alpha$ are typically used as the X-ray sources for XPS. Do they excite 1s photoelectron emission of metals such as Ti and Fe? If not, how do we detect such elements by XPS? (20 points)